

Fair representation by matchings in bipartite graphs:

Conjecture 1.15 of Aharoni, Alon, Berger, Chudnovsky, Kotlar, Loebl and Ziv

holds with $c(m) = 32(m + 1)^3$

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Machine-generated note

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Abstract

Aharoni, Alon, Berger, Chudnovsky, Kotlar, Loebl and Ziv conjectured (Conjecture 1.15 of arXiv:1611.03196) that for every m there is a constant $c(m)$ such that, for every bipartite graph G with maximum degree Δ and arbitrary edge sets $E_1, \dots, E_m \subseteq E(G)$, there is a matching S with $|S| \geq |E(G)|/\Delta - c(m)$ and $|S \cap E_i| \leq \lceil |E_i|/\Delta \rceil$ for all i . We prove the conjecture with the explicit, unoptimised constant $c(m) = 32(m + 1)^3$. The proof writes the target vector $(|E(G)|/\Delta, |E_1|/\Delta, \dots, |E_m|/\Delta)$ as a convex combination of the statistics of at most $m + 2$ matchings (König plus Carathéodory) and then shows, using the Stromquist–Woodall interval-splitting theorem on the alternating cycles of two matchings, that a convex combination of two matchings’ statistics can be realised by a single matching up to an additive error $32(m + 1)$. The authors’ suggested sharp value $c(m) = m/2$ remains open.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and was subsequently checked by human mathematicians. [Section 5](#) records the full provenance.

1 Introduction

All graphs are finite. For a graph G let $\Delta(G)$ be its maximum degree; a *matching* is a set of pairwise disjoint edges. In their study of fair representation, Aharoni, Alon, Berger, Chudnovsky, Kotlar, Loebl and Ziv [1] observe that “over-representation” statements fail when the sets E_i do not partition $E(G)$, and propose the following under-representation formulation.

Conjecture 1.1 ([1, Conjecture 1.15]). For every m there exists a number $c(m)$ for which the following is true: if G is a bipartite graph and $E_1, \dots, E_m$ are any sets of edges, then there exists a matching S in G of size at least $\frac{|E(G)|}{\Delta(G)} - c(m)$ such that $|S \cap E_i| \leq \lceil \frac{|E_i|}{\Delta(G)} \rceil$ for all $i \leq m$.

The sets E_i are arbitrary subsets of $E(G)$: no disjointness is assumed. The authors suggest that $c(m) = m/2$ may suffice. We prove the conjecture with an explicit constant.

Theorem 1.2 (Main theorem). For every $m \geq 1$, [Conjecture 1.1](#) holds with

$$c(m) = 32(m + 1)^3.$$

That is, if G is a bipartite graph with $D = \Delta(G) \geq 1$ and $E_1, \dots, E_m \subseteq E(G)$, then G has a matching S with

$$|S| \geq \frac{|E(G)|}{D} - 32(m + 1)^3 \quad \text{and} \quad |S \cap E_i| \leq \left\lceil \frac{|E_i|}{D} \right\rceil \quad (1 \leq i \leq m).$$

The constant is deliberately not optimised; see [Remark 4.1](#).

Idea of proof. Attach to a matching M its *statistic vector* $z(M) = (|M|, |M \cap E_1|, \dots, |M \cap E_m|) \in \mathbb{R}^{m+1}$. By König’s line-colouring theorem $E(G)$ is the disjoint union of D matchings, so the target $p = (|E(G)|/D, |E_1|/D, \dots, |E_m|/D)$ is the average of their statistic vectors; by Carathéodory’s theorem p is a convex combination of the statistic vectors of at most $m + 2$ matchings ([Lemma 3.1](#)). The heart of the argument is an *interpolation lemma* ([Lemma 2.2](#)): for two matchings A, B and $\theta \in [0, 1]$ there is a matching C whose statistics are, coordinate by coordinate, within $32(m + 1)$ of $\theta z(A) + (1 - \theta)z(B)$, the size from below and the counts $|C \cap E_i|$ from above. It is proved by extending A, B to perfect matchings, splitting their alternating cycles into unit cells (a_j, b_j) , choosing by the Stromquist–Woodall theorem a union of few intervals of cells carrying exactly a θ -share of every statistic, taking the A -edges on those cells and the B -edges elsewhere, and deleting one edge at each of the few places where the choice switches. Iterating the interpolation along the convex combination accumulates an error at most $32(m + 1)^2$ ([Lemma 3.2](#)); finally at most $32(m + 1)^2$ edges per set E_i are deleted to turn the upper bounds $|E_i|/D + 32(m + 1)^2$ into the exact ceilings, losing $32(m + 1)^3$ in total.

2 Two ingredients

2.1 Proportional interval splitting

The only external ingredient is the following consequence of the theorem of Stromquist and Woodall [[3](#)]: for r nonatomic probability measures on $[0, 1]$ and any $\theta \in [0, 1]$ there is a union of at most r intervals that has measure exactly θ for each of them. We use it in the following weaker form (non-probability measures are normalised, zero measures impose no condition, and any compact interval is rescaled to $[0, 1]$).

Lemma 2.1 (Stromquist–Woodall, weak form). *Let $\nu_1, \dots, \nu_r$ be nonatomic finite nonnegative measures on a compact interval I and let $0 \leq \theta \leq 1$. There is a set $X \subseteq I$ which is a union of at most $2r$ intervals such that*

$$\nu_j(X) = \theta \nu_j(I) \quad (1 \leq j \leq r).$$

We apply it only to measures with piecewise-constant densities. The bound $2r$ is what the writeup uses; the theorem gives r .

2.2 Interpolating two matchings

For a bipartite graph H , a *labelling* assigns to each edge e a vector $w(e) = (w_0(e), \dots, w_{d-1}(e)) \in \{0, 1\}^d$; for a set M of edges write $w_\ell(M) = \sum_{e \in M} w_\ell(e)$.

Lemma 2.2 (Interpolation lemma). *Let H be a bipartite graph with a labelling $w : E(H) \rightarrow \{0, 1\}^d$, let A, B be matchings in H and $0 \leq \theta \leq 1$. There is a matching C in H such that*

$$w_0(C) \geq \theta w_0(A) + (1 - \theta)w_0(B) - 32d$$

and, for $1 \leq j < d$,

$$w_j(C) \leq \theta w_j(A) + (1 - \theta)w_j(B) + 32d.$$

Idea of proof. Extend A and B by dummy vertices and zero-labelled dummy edges to perfect matchings A^*, B^* of one balanced bipartite graph. Their symmetric difference is a disjoint union of even alternating cycles $a_1 b_1 a_2 b_2 \dots a_s b_s$ ($a_j \in A^*, b_j \in B^*$); call (a_j, b_j) a *cell* and lay all cells out as unit subintervals of one interval. The $2d$ measures “label ℓ of a_j ” and “label ℓ of b_j ” spread uniformly over cell j are nonatomic, so [Lemma 2.1](#) gives a union X of at most $4d$ intervals carrying exactly a θ -share of each. Let J be the set of cells whose midpoint lies in X ; at most $8d$ cells are cut by the boundary of X , so

the A -edges of J carry a θ -share of every label up to $8d$, and the B -edges outside J a $(1 - \theta)$ -share up to $8d$. Select a_j for $j \in J$, b_j for $j \notin J$, and all common edges; this set has the right statistics up to $16d$ but fails to be a matching exactly where a selected b_{j-1} meets a selected a_j . The indicator of J switches at most $8d$ times along the line and at most $16d$ times cyclically, so deleting a_j at every such switch removes at most $16d$ edges and yields a matching; discarding dummies costs nothing since they carry label 0. [\[full proof\]](#)

3 Proof of the main theorem

Fix G , $D = \Delta(G) \geq 1$ and $E_1, \dots, E_m \subseteq E(G)$ as in [Theorem 1.2](#). Put

$$d = m + 1, \quad z(M) = (|M|, |M \cap E_1|, \dots, |M \cap E_m|) \in \mathbb{R}^d$$

for every matching M of G , and label the edges of G by

$$w_0(e) = 1, \quad w_i(e) = \mathbf{1}_{\{e \in E_i\}} \quad (1 \leq i \leq m), \quad (1)$$

so that $w_\ell(M)$ is the ℓ -th coordinate of $z(M)$ for every matching M and $0 \leq \ell \leq m$.

Lemma 3.1 (Reduction to $m + 2$ matchings). *There are matchings $M_1, \dots, M_k$ of G with $k \leq d + 1 = m + 2$ and positive reals $\lambda_1, \dots, \lambda_k$ with $\sum_j \lambda_j = 1$ such that*

$$p := \left(\frac{|E(G)|}{D}, \frac{|E_1|}{D}, \dots, \frac{|E_m|}{D} \right) = \sum_{j=1}^k \lambda_j z(M_j). \quad (2)$$

Idea of proof. König's line-colouring theorem partitions $E(G)$ into D matchings, whose statistic vectors average to p ; Carathéodory's theorem in $\mathbb{R}^d$ selects at most $d + 1$ of them. [\[full proof\]](#)

Lemma 3.2 (Iterated interpolation). *With $M_1, \dots, M_k$ and $\lambda_1, \dots, \lambda_k$ as in [Lemma 3.1](#), there is a matching C of G with*

$$|C| \geq \frac{|E(G)|}{D} - 32d^2 \quad \text{and} \quad |C \cap E_i| \leq \frac{|E_i|}{D} + 32d^2 \quad (1 \leq i \leq m). \quad (3)$$

Idea of proof. Let $\Lambda_j = \lambda_1 + \dots + \lambda_j$ and $p_j = \Lambda_j^{-1} \sum_{\ell \leq j} \lambda_\ell z(M_\ell)$. Start with $C_1 = M_1$ and obtain C_j from C_{j-1} and M_j by [Lemma 2.2](#) with $\theta = \Lambda_{j-1}/\Lambda_j$ and the labels (1). Since $p_j = \theta p_{j-1} + (1 - \theta)z(M_j)$ and $\theta \leq 1$, the error after j steps is at most $(j - 1) \cdot 32d$; as $k - 1 \leq d$ and $p_k = p$, this gives (3). [\[full proof\]](#)

Proof of [Theorem 1.2](#). Let C be the matching of [Lemma 3.2](#), and put $q_i = \lceil |E_i|/D \rceil$ and $B = 32d^2$. By (3) and $|E_i|/D \leq q_i$,

$$|C \cap E_i| - q_i \leq B \quad (1 \leq i \leq m).$$

Process $i = 1, \dots, m$ in turn: while the current matching contains more than q_i edges of E_i , delete an arbitrary edge of E_i from it. A subset of a matching is a matching; deletions never increase any count $|\cdot \cap E_{i'}|$, so the bound $|\cdot \cap E_{i'}| \leq q_{i'}$ obtained for earlier $i' < i$ persists (this is why overlapping sets E_i cause no difficulty); and at most B edges are deleted while processing i , hence at most mB in total. The resulting matching S satisfies $|S \cap E_i| \leq q_i = \lceil |E_i|/D \rceil$ for every i , and by (3)

$$|S| \geq |C| - mB \geq \frac{|E(G)|}{D} - B - mB = \frac{|E(G)|}{D} - 32d^2(m + 1) = \frac{|E(G)|}{D} - 32(m + 1)^3. \quad \square$$

4 Remarks

Remark 4.1 (The constant). The value $32(m+1)^3$ is very crude and no attempt was made to optimise it. The proof gives $16d$ rather than $32d$ for the upper bounds in [Lemma 2.2](#), and [Lemma 2.1](#) holds with r intervals instead of $2r$; neither improvement was used. The authors’ suggested value $c(m) = m/2$ [1] is not approached by this argument and remains open. The referee’s brute-force search over 587 small instances ([Appendix B](#)) found a maximum deficit $|E(G)|/D - \max |S|$ of $\frac{1}{2}$, consistent with $c(m) = m/2$.

Remark 4.2 (Slack recorded by the referee). The referee notes two places where the writeup states a weaker bound than it proves: [Lemma 2.2](#) asserts $+32d$ for the coordinates $j \geq 1$ where the proof yields $+16d$, and [Lemma 2.1](#) is stated with $2r$ intervals where Stromquist–Woodall give r . Both are deliberate weakenings, not errors; we keep the writeup’s statements.

Remark 4.3 (Conventions and edge cases). The referee records that $D = \Delta(G) \geq 1$ is assumed (for an edgeless graph the quotient $|E(G)|/\Delta(G)$ is undefined and the conjecture is vacuous), and that the E_i are sets of edges of G (allowing non-edges in E_i would only weaken the claim). The argument works verbatim for bipartite multigraphs, since König’s line-colouring theorem holds for them. The proof is non-constructive at exactly one point, the topological existence statement of [Lemma 2.1](#); existence is all the conjecture asks for.

Remark 4.4 (Computational checks). The referee ([Appendix B](#)) implemented the dummy extension, the cell decomposition, the selection and the conflict repair of [Lemma 2.2](#) and verified on 4000 random bipartite instances that the repaired selection is always a matching, that the number of deletions is at most $16d$ for interval-structured J , and that the pre-repair statistic identity holds; it also ran the whole pipeline of [Section 3](#) on bipartite circulant graphs with exact rational Carathéodory reduction (always $k \leq m+2$), observing worst per-step interpolation errors of 8 against the allowed $32d \geq 64$ and all ceilings satisfied. The one step code cannot certify is the exactness of the Stromquist–Woodall split, which the script replaced by a numerical search.

Remark 4.5 (Novelty). The referee’s searches found no resolution of [Conjecture 1.1](#) in the literature. The closest work it located is Othman and Berger [2], which treats perfect matchings of $K_{n,n}$ under partitions with a two-sided fairness measure and reports an unpublished 2014 result for $K_{n,n}$ with partitions; this is strictly narrower than [Conjecture 1.1](#) (complete bipartite graphs only, partitions only). The interpolation-between-matchings idea is folklore-adjacent, so an unindexed prior proof cannot be excluded. A subsequent human review checked the mathematical proof and reported no error; it did not establish literature priority.

5 Provenance

The proof was generated, machine-refereed and rewritten without human intervention, in three stages. (1) The proof was found by the language model gpt-5.6-sol in a single autonomous pass on 2026-09-01, as part of a large sweep over open problems catalogued from arXiv (catalog id 1611.03196__03; original writeup in `attacks/1611.03196__03/output.md`). (2) An independent adversarial referee agent (claude-fable-5) re-derived every step, checked the conjecture verbatim against the arXiv source and the Stromquist–Woodall reference, and machine-tested the combinatorial core of the interpolation lemma on 4000 random instances, the statement itself by brute force on 587 small instances, and the whole pipeline end to end on bipartite circulant graphs; its verdict was CONFIRMED. Its report is reproduced verbatim in [Appendix B](#). (3) On 2026-09-09 the writeup was rewritten into the present note by claude-fable-5.1: the text was reorganised with full proofs in [Appendix A](#), the interval-splitting theorem is now cited precisely, and steps that the writeup stated in one line (the rounding error, the transition count, the induction) are spelled out exactly as the referee re-derived them. Two observations of the referee are recorded as remarks and marked as such ([Remarks 4.2](#) and [4.3](#)). No mathematical content was changed. (4) On 2026-09-18, the proof was checked by human mathematicians, who

reported no mathematical error. This review changed only the review-status disclosures in the note; the proof was left unchanged.

A Full proofs

Lemma 2.2 (Interpolation lemma). *Let H be a bipartite graph with a labelling $w : E(H) \rightarrow \{0, 1\}^d$, let A, B be matchings in H and $0 \leq \theta \leq 1$. There is a matching C in H such that*

$$w_0(C) \geq \theta w_0(A) + (1 - \theta)w_0(B) - 32d$$

and, for $1 \leq j < d$,

$$w_j(C) \leq \theta w_j(A) + (1 - \theta)w_j(B) + 32d.$$

Proof. Let $L \cup R$ be the bipartition of H , $|L| = p$, $|R| = q$.

Dummy extension. Add q new (dummy) vertices to the left side and p dummy vertices to the right side, and add every edge between a dummy vertex and a vertex of the opposite side. The resulting bipartite graph H^* has $p + q$ vertices on each side. Give every added edge the label $0 \in \{0, 1\}^d$, so that $w_\ell(M)$ is unchanged for every $M \subseteq E(H)$. Every matching M of H extends to a perfect matching M^* of H^* : match the $p - |M|$ unmatched original left vertices to distinct dummy right vertices (there are p of them), match the $q - |M|$ unmatched original right vertices to distinct dummy left vertices (there are q), and match the remaining $|M|$ dummy right vertices bijectively to the remaining $|M|$ dummy left vertices. Do this for A and B , obtaining perfect matchings A^*, B^* of H^* with $w_\ell(A^*) = w_\ell(A)$ and $w_\ell(B^*) = w_\ell(B)$ for all ℓ .

Cells. In $A^* \triangle B^*$ every vertex has degree 0 or 2 (one edge of each perfect matching), so the nontrivial components are cycles alternating between A^* and B^* , hence even. On each such cycle write the edges in cyclic order as $a_1, b_1, a_2, b_2, \dots, a_s, b_s$ with $a_j \in A^*$, $b_j \in B^*$; thus a_j shares one end with b_j and its other end with b_{j-1} (indices modulo s within the cycle), and these are the only incidences among the edges of the cycle. Call (a_j, b_j) the j -th cell of the cycle. Fix an order of the cycles, concatenate their cell lists into one list of n cells, and let cell number j ($1 \leq j \leq n$) correspond to the unit interval $[j - 1, j] \subseteq I := [0, n]$.

Splitting. For $0 \leq \ell < d$ define measures ν_ℓ^A, ν_ℓ^B on I by giving the interval of cell j constant density $w_\ell(a_j)$, respectively $w_\ell(b_j)$. These $2d$ measures are finite, nonnegative and nonatomic, and $\nu_\ell^A(I) = \sum_j w_\ell(a_j)$, $\nu_\ell^B(I) = \sum_j w_\ell(b_j)$. By Lemma 2.1 with $r = 2d$ there is a set $X \subseteq I$, a union of at most $4d$ intervals, with

$$\nu_\ell^A(X) = \theta \nu_\ell^A(I), \quad \nu_\ell^B(X) = \theta \nu_\ell^B(I) \quad (0 \leq \ell < d).$$

X has at most $8d$ boundary points.

Rounding to whole cells. Let J be the set of cells whose midpoint $j - \frac{1}{2}$ lies in X . A cell whose interval contains no boundary point of X in its interior is either contained in X (then it lies in J and contributes its full mass to $\nu_\ell^A(X)$) or disjoint from the interior of X (then it is not in J and contributes nothing). A cell cut by a boundary point contributes at most its total mass $w_\ell(a_j) \leq 1$ to either side. Since at most $8d$ cells are cut,

$$\left| \sum_{j \in J} w_\ell(a_j) - \theta \sum_j w_\ell(a_j) \right| = \left| \sum_{j \in J} w_\ell(a_j) - \nu_\ell^A(X) \right| \leq 8d, \quad (4)$$

and in the same way

$$\left| \sum_{j \in J} w_\ell(b_j) - \theta \sum_j w_\ell(b_j) \right| \leq 8d. \quad (5)$$

Selection. Let

$$T = (A^* \cap B^*) \cup \{a_j : j \in J\} \cup \{b_j : j \notin J\}.$$

Then $w_\ell(T) = w_\ell(A^* \cap B^*) + \sum_{j \in J} w_\ell(a_j) + \sum_{j \notin J} w_\ell(b_j)$, while

$$\theta w_\ell(A^*) + (1 - \theta)w_\ell(B^*) = w_\ell(A^* \cap B^*) + \theta \sum_j w_\ell(a_j) + (1 - \theta) \sum_j w_\ell(b_j).$$

Subtracting, and using $\sum_{j \notin J} w_\ell(b_j) - (1 - \theta) \sum_j w_\ell(b_j) = -(\sum_{j \in J} w_\ell(b_j) - \theta \sum_j w_\ell(b_j))$, the bounds (4) and (5) give

$$|w_\ell(T) - (\theta w_\ell(A^*) + (1 - \theta)w_\ell(B^*))| \leq 16d \quad (0 \leq \ell < d). \quad (6)$$

Conflicts. Vertices on edges of $A^* \cap B^*$ lie on no cycle, so they are covered exactly once by T . A vertex on a cycle is the common end of a_j and b_j for some j , or of b_{j-1} and a_j . In the first case it is covered exactly once, since exactly one of a_j, b_j is selected. In the second case it is covered twice if and only if b_{j-1} and a_j are both selected, i.e. $j - 1 \notin J$ and $j \in J$ (cyclically within the cycle); call such a j an *up-transition*. Delete a_j from T for every up-transition j . Afterwards every vertex is covered at most once (the other end of a deleted a_j is the common end with b_j , which is not selected), so the result T' is a matching of H^* .

Counting deletions. Read the indicator of J along the linear order of the n cells. If it differs on consecutive cells $j, j + 1$, the midpoints $j - \frac{1}{2}$ and $j + \frac{1}{2}$ lie on different sides of X , so X has a boundary point in the open interval $(j - \frac{1}{2}, j + \frac{1}{2})$; these intervals are disjoint for distinct j , hence the indicator changes at most $8d$ times along the line. Within one cycle, the cyclic changes are its internal changes (those between consecutive cells of the cycle in the linear order) plus possibly one change between its last and first cell; the number of cyclic changes of a cyclic 0/1 sequence is even, so a wrap-around change can only occur if the cycle has at least one internal change. Hence the total number of cyclic changes is at most twice the number of internal changes, i.e. at most $16d$. Every up-transition is a cyclic change, so at most $16d$ edges are deleted.

Conclusion. Let $C = T' \cap E(H)$, a matching of H . Dummy edges carry label 0, so $w_\ell(C) = w_\ell(T')$ for all ℓ . Each deleted edge decreases w_ℓ by at most 1, and deletions never increase any w_ℓ . Therefore, by (6) and $w_\ell(A^*) = w_\ell(A)$, $w_\ell(B^*) = w_\ell(B)$,

$$w_j(C) \leq w_j(T) \leq \theta w_j(A) + (1 - \theta)w_j(B) + 16d \leq \theta w_j(A) + (1 - \theta)w_j(B) + 32d \quad (1 \leq j < d),$$

and

$$w_0(C) \geq w_0(T) - 16d \geq \theta w_0(A) + (1 - \theta)w_0(B) - 16d - 16d = \theta w_0(A) + (1 - \theta)w_0(B) - 32d. \quad \square$$

Lemma 3.1 (Reduction to $m + 2$ matchings). *There are matchings $M_1, \dots, M_k$ of G with $k \leq d + 1 = m + 2$ and positive reals $\lambda_1, \dots, \lambda_k$ with $\sum_j \lambda_j = 1$ such that*

$$p := \left(\frac{|E(G)|}{D}, \frac{|E_1|}{D}, \dots, \frac{|E_m|}{D} \right) = \sum_{j=1}^k \lambda_j z(M_j). \quad (2)$$

Proof. By König's line-colouring theorem, the edge set of the bipartite graph G can be partitioned into $D = \Delta(G)$ matchings $M'_1, \dots, M'_D$. Since the M'_j partition $E(G)$, we have $\sum_{j=1}^D |M'_j| = |E(G)|$ and $\sum_{j=1}^D |M'_j \cap E_i| = |E_i|$ for each i , i.e.

$$p = \frac{1}{D} \sum_{j=1}^D z(M'_j).$$

Thus p lies in the convex hull of the finite set $\{z(M'_1), \dots, z(M'_D)\} \subseteq \mathbb{R}^d$. By Carathéodory's theorem, p is a convex combination of at most $d + 1$ of these points; discarding zero coefficients we obtain matchings $M_1, \dots, M_k$, $k \leq d + 1 = m + 2$, and positive $\lambda_1, \dots, \lambda_k$ with $\sum_j \lambda_j = 1$ satisfying (2). $\square$

Lemma 3.2 (Iterated interpolation). *With $M_1, \dots, M_k$ and $\lambda_1, \dots, \lambda_k$ as in Lemma 3.1, there is a matching C of G with*

$$|C| \geq \frac{|E(G)|}{D} - 32d^2 \quad \text{and} \quad |C \cap E_i| \leq \frac{|E_i|}{D} + 32d^2 \quad (1 \leq i \leq m). \quad (3)$$

Proof. For $1 \leq j \leq k$ let $\Lambda_j = \sum_{\ell=1}^j \lambda_\ell$ (so $\Lambda_k = 1$ and $\Lambda_j > 0$) and

$$p_j = \frac{1}{\Lambda_j} \sum_{\ell=1}^j \lambda_\ell z(M_\ell),$$

so that $p_1 = z(M_1)$ and $p_k = p$ by (2). For $j \geq 2$ put $\theta_j = \Lambda_{j-1}/\Lambda_j \in (0, 1)$; then $1 - \theta_j = \lambda_j/\Lambda_j$ and

$$p_j = \theta_j p_{j-1} + (1 - \theta_j) z(M_j). \quad (7)$$

Define matchings $C_1, \dots, C_k$ of G by $C_1 = M_1$ and, for $j \geq 2$, by applying Lemma 2.2 to $H = G$ with the labelling (1) (so $d = m + 1$), $A = C_{j-1}$, $B = M_j$ and $\theta = \theta_j$; let C_j be the matching it produces. We claim that for $1 \leq j \leq k$ and $1 \leq i \leq m$,

$$|C_j| \geq (p_j)_0 - (j - 1) 32d, \quad |C_j \cap E_i| \leq (p_j)_i + (j - 1) 32d. \quad (8)$$

For $j = 1$ both hold with equality since $z(C_1) = p_1$. Let $j \geq 2$ and assume (8) for $j - 1$. With the labels (1), $w_0(M) = |M|$ and $w_i(M) = |M \cap E_i|$, so Lemma 2.2, the induction hypothesis, $\theta_j \leq 1$ and (7) give

$$\begin{aligned} |C_j| &\geq \theta_j |C_{j-1}| + (1 - \theta_j) |M_j| - 32d \\ &\geq \theta_j ((p_{j-1})_0 - (j - 2) 32d) + (1 - \theta_j) z(M_j)_0 - 32d \geq (p_j)_0 - (j - 1) 32d, \end{aligned}$$

and likewise

$$\begin{aligned} |C_j \cap E_i| &\leq \theta_j |C_{j-1} \cap E_i| + (1 - \theta_j) |M_j \cap E_i| + 32d \\ &\leq \theta_j ((p_{j-1})_i + (j - 2) 32d) + (1 - \theta_j) z(M_j)_i + 32d \leq (p_j)_i + (j - 1) 32d. \end{aligned}$$

This proves (8). Taking $j = k$, using $p_k = p$ and $k - 1 \leq d$ (from Lemma 3.1), the matching $C = C_k$ satisfies

$$|C| \geq \frac{|E(G)|}{D} - d \cdot 32d = \frac{|E(G)|}{D} - 32d^2, \quad |C \cap E_i| \leq \frac{|E_i|}{D} + 32d^2,$$

which is (3). □

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-fable-5, 2026-09-02) on the *original* writeup. Its step numbers refer to that writeup, not to this note. The scripts it mentions are in `verification/scripts/1611.03196__03/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The catalog statement was checked verbatim against arXiv:1611.03196v1 (fetched via the arXiv HTML rendering). Conjecture 1.15 there reads exactly as in `attacks/1611.03196__03/prompt.md`: for every m there is $c(m)$ such that for any bipartite G and **arbitrary** edge sets $E_1, \dots, E_m$ (the paper says “any sets of edges” — no disjointness or partition requirement; I confirmed this from the source text and its surrounding paragraphs, which explicitly motivate the under-representation formulation because the over-representation versions fail for non-partitions), G has a matching S with $|S| \geq |E(G)|/\Delta(G)$

- $c(m)$ and $|S \cap E_i| \leq \text{ceil}(|E_i|/\Delta(G))$ for all i . The writeup formalizes precisely this and proves the existence of $c(m)$ with the explicit value $32(m+1)^3$. This is the intended reading: the conjecture asks only for existence of some $c(m)$; the authors' further suggestion that $c(m) = m/2$ may suffice is a stronger refinement, which the writeup explicitly leaves open (both in its verdict block and in its Remarks). A reasonable author of the original paper would consider Conjecture 1.15 as stated resolved by this result, though the sharp $m/2$ question remains. No interpretation gaming detected. (Note: in the later journal/author version of the paper, e.g. fairrep4.pdf, the numbering shifts and "Theorem 1.15" is a different result about paths; the catalog id refers to the arXiv v1 numbering, which matches.)

B.2 Step-by-step findings

step	label	note
1. Lemma 1 (proportional interval splitting: r nonatomic measures, share θ , X a union of $\leq 2r$ intervals with $\nu_j(X) = \theta \nu_j(I)$)	VALID	This is a weakening of the Stromquist-Woodall theorem (J. Math. Anal. Appl. 108 (1985) 241-248), which gives $\leq r$ intervals on $[0,1]$ for r nonatomic probability measures and any θ . Non-probability measures normalize (zero measures are trivially split), and rescaling $[0,1]$ to the cell interval is harmless. The writeup's parenthetical "the usual stronger statement gives at most r intervals" is accurate. Reference confirmed (see below). With
2. Lemma 2, dummy extension (both matchings A, B extend to perfect matchings A, B of one balanced bipartite supergraph; dummy edges labeled 0)	VALID	
3. Lemma 2, cycle/cell structure (A^* triangle B^* is a disjoint union of even alternating cycles $a_1 b_1 \dots a_s b_s$; a_j meets exactly b_{j-1} and b_j)	VALID	Standard for two perfect matchings (no paths can occur). The adjacency pattern (a_j shares a vertex with b_j and with b_{j-1} , cyclically) was verified on every cycle of every random instance.
4. Lemma 2, application of Lemma 1 to the $2d$ cell measures, giving $X =$ union of $\leq 4d$ intervals with exact θ -shares	VALID	$2d$ measures with piecewise-constant $\{0,1\}$ -densities on unit cells are nonatomic; Lemma 1 with $r = 2d$ gives $\leq 4d$ intervals (Stromquist-Woodall actually gives $\leq 2d$).

step	label	note
5. Lemma 2, midpoint rounding error $\leq 8d$ per measure	VALID	X has $\leq 8d$ boundary points; only cells properly cut by a boundary point contribute error, each at most the cell's mass ≤ 1 (labels are in $\{0,1\}$); a boundary point at an integer cell boundary cuts no cell. Re-derived: $\leq 16d$
6. Lemma 2, pre-repair bound (1):	$w_l(T) - (\theta w_l(A) + (1-\theta)w_l(B))$	
7. Lemma 2, conflict analysis and repair (only conflicts are b_{j-1} , a_j at cyclic up-transitions $j-1 \notin J$, $j \in J$; deleting a_j at each yields a matching)	VALID	Per cell exactly one of a_j , b_j is selected, so the only doubly-covered vertices are those between b_{j-1} and a_j at up-transitions; each selected a_j has at most one conflict. Verified computationally: after repair, the edge set (plus common edges) is a matching in every one of 4000 random instances with both random and interval-structured J (check C2).
8. Lemma 2, deletion count $\leq 16d$	VALID	The sampled indicator changes $\leq 8d$ times linearly (one X-boundary point per change); each cycle whose cyclic transition count exceeds its internal changes consumes at least one internal change, so cyclic transitions $\leq 8d + 8d = 16d$, and up-transitions (= deletions) are at most that. Verified computationally in interval mode (check C3).
9. Lemma 2, conclusion ($w_0(C) \geq \theta w_0(A) + (1-\theta)w_0(B) - 32d$; $w_j(C) \leq \dots + 32d$ for $j \geq 1$)	VALID	Each deleted edge lowers w_0 by ≤ 1 , so w_0 loses $\leq 16d$ beyond (1); side coordinates only decrease under deletion (the proven $+16d$ is stronger than the stated $+32d$). Dummy edges have zero labels, so $w_l(A^*) = w_l(A)$ and discarding dummies from C changes nothing.

step	label	note
10. Section 3, Koenig decomposition and p in the convex hull	VALID	$\chi'(G) = \Delta(G)$ for bipartite G (Koenig's line-coloring theorem), so $E(G)$ partitions into D matchings; summing statistic vectors gives $p = (1/D) * \sum z(M_j')$ exactly (partition $\Rightarrow$ sum
11. Section 3, Caratheodory reduction to $k \leq m+2$ matchings	VALID	Caratheodory in $\mathbb{R}^d$, $d = m+1$: p is a convex combination of $\leq d+1 = m+2$ of the $z(M_j')$. Implemented exactly over rationals in the end-to-end test; $k \leq d+1$ always held and the point p was preserved.
12. Section 4, iterated interpolation and the induction error $(j-1)*32d$	VALID	With $\theta = \lambda_{j-1}/\lambda_j$ one has $p_j = \theta p_{j-1} + (1-\theta) z(M_j)$ (since $\lambda_j/\lambda_j = 1-\theta$). Re-derived: $\text{error}_j \leq \theta(j-2)32d + 32d \leq (j-1)*32d$, with base case $\text{error}_1 = 0$.
13. Section 4, bounds (3)-(4) with $32d^2$	VALID	$k-1 \leq d$, so the accumulated error is $\leq d * 32d = 32d^2$; $p_k = p$, whose coordinates are
14. Section 5, ceiling enforcement (delete $\leq B$ edges per set, $\leq mB$ total; upper bounds preserved)	VALID	
15. Section 5, final arithmetic $c(m) = 32(m+1)^3$	VALID	$B + mB = (m+1) * 32(m+1)^2 = 32(m+1)^3$. Verified.

No GAP or ERROR labels. The one presentational looseness (Lemma 2 states $+32d$ for side coordinates where $+16d$ is proven, and Lemma 1 states $2r$ intervals where r suffices) is a deliberate weakening, not an error.

B.3 Reference check

- **Conjecture 1.15, arXiv:1611.03196v1** (Aharoni, Alon, Berger, Chudnovsky, Kotlar, Loebl, Ziv, "Fair representation by independent sets"): fetched via <https://arxiv.org/html/1611.03196v1>. The conjecture, its "any sets of edges" hypothesis, the $m/2$ suggestion, and the surrounding under-representation motivation all match the prompt and the writeup's formalization verbatim.
- **Stromquist-Woodall, "Sets on which several measures agree", JMAA 108 (1985) 241-248**: confirmed (ScienceDirect listing plus secondary sources, e.g. the Crew-Narayanan-Spirkl "Disproportionate division" paper and Goldberg et al. "Consensus division in an arbitrary ratio", ITCS 2023, which state it in exactly this form): for n nonatomic probability measures on $[0,1]$ and any α in $[0,1]$ there is a union of at most n intervals with measure α in every measure. This covers Lemma 1 (which asks only for $2r$ intervals).

- **Koenig’s line-coloring theorem** ($\chi' = \Delta$ for bipartite graphs) and **Caratheodory’s theorem**: textbook results, used with correct hypotheses.
- **Novelty / ALREADY_KNOWN check**: targeted searches (“Conjecture 1.15” + fair representation; under-representation matching $|E|/\Delta$) found no paper resolving Conjecture 1.15. The closest literature is Othman-Berger, “Almost fair perfect matchings in complete bipartite graphs”, Discrete Mathematics 347 (2024) (the paper the catalog reviewer could not retrieve): it concerns perfect matchings of $K_{\{n,n\}}$ under partitions with a two-sided fairness spread $s(M)$, proves $m = 2, 3$ and a weak $m = 4$ bound, and cites an unpublished 2014 result giving a $K_{\{n,n\}}$ -partition analogue with a constant $f(m)$ — related in spirit but strictly narrower than Conjecture 1.15 (complete bipartite only, partitions only). The catalog’s own auto-review (May 2026) also found no resolution. The conjecture appears to be genuinely open in the literature.

B.4 Computational check

Scripts in `verification/scripts/1611.03196__03/`:

1. `test_lemma2_construction.py` — the combinatorial core of Lemma 2. 4000 random bipartite instances (sides up to 9, all densities), random matchings A, B , the writeup’s exact dummy extension, cycle/cell decomposition, and both arbitrary and interval-induced cell subsets J . Verified: extensions are perfect matchings; cycles alternate with the claimed adjacency pattern; after deleting a_j at every cyclic up-transition the selection plus common edges is always a matching; deletions $\leq$ cyclic transitions $\leq 16d$ for interval-structured X ; the pre-repair statistic identity holds for random $\{0,1\}^d$ labels. Result: **all 4000 instances passed all checks**.
2. `test_conjecture_smallcases.py` — statement-level sanity. 587 random instances ($|E| \leq 22$, $m \leq 3$, subsets/partitions/overlapping families), brute-force branch-and-bound maximization of $|S|$ subject to the ceiling caps. Result: **maximum observed deficit $|E|/\Delta - \max|S|$ was $1/2$** (at $|E| = 5$, $\Delta = 2$, $m = 2$), consistent with the conjecture, with the authors’ suggested $c(m) = m/2$, and far inside the writeup’s $32(m+1)^3$.
3. `test_pipeline_end2end.py` — the full Sections 3-5 pipeline on bipartite circulant graphs (n up to 24, D up to 24, m in $\{1,2,3\}$) with explicit Koenig decompositions: exact rational Caratheodory reduction ($k \leq m+2$ held in all trials, point preserved exactly), iterated Lemma-2 interpolation using a coordinate-descent search for the union-of-intervals split (per-measure split error ≤ 0.30 achieved; the exactness of the ideal split is the Stromquist-Woodall citation, which is the one step code cannot certify), conflict repair, and the final ceiling-enforcement deletions. Result: **every intermediate C_j and the final S was a valid matching; worst per-step interpolation error observed was 8 against the allowed $32d \geq 64$; final deficits were at most 11 against allowed bounds 256-2048; all caps $|S \cap E_i| \leq \text{ceil}(|E_i|/D)$ held**.

No computational check failed.

B.5 Caveats

- **$\Delta(G) \geq 1$ is implicitly assumed** (the writeup states $D \geq 1$): for an edgeless graph $|E|/\Delta$ is $0/0$; the conjecture is vacuous there under any reasonable convention. Not a real gap.
- **$E_i \subseteq E(G)$ is implicitly assumed**. The source paper says “any sets of edges” (of G); if one perversely allowed E_i to contain non-edges the claim only gets weaker (larger ceiling, same intersections). Harmless.
- **The proof is non-constructive** at exactly one point: the Stromquist-Woodall split is topological (Borsuk-Ulam-type); the end-to-end script therefore uses a numerical search as a stand-in for that one step. Existence, which is all the conjecture asks, is unaffected.

- **Lemma 1 is stated slightly informally** (“standard form of the proportional continuous necklace-splitting theorem”) without naming Stromquist-Woodall; a journal version should cite it precisely. The stated $2r$ bound is safe (true bound r).
- **The sharp question $c(m) = m/2$ remains open**, as the writeup itself says; the verdict block’s one-line claim matches what is actually proven (existence of $c(m)$, with $32(m+1)^3$).
- **Novelty is probable but not certain**: the interpolation-between-matchings trick is folklore-adjacent (cf. the unpublished 2014 $K_{\{n,n\}}$ result reported in Othman-Berger 2024), so an undindexed prior proof cannot be fully excluded; the writeup flags this honestly. All searches performed found no prior resolution.
- The multigraph remark (Remark 1) is correct (Koenig’s line-coloring theorem holds for bipartite multigraphs) but was not needed for the verdict.

B.6 Referee summary

I approached this as a hostile referee and failed to break the proof. Every step re-derives cleanly: the dummy extension, alternating-cycle cell structure, midpoint rounding against an exact Stromquist-Woodall split, the conflict-repair bound of $16d$ deletions, the theta-weighted induction through at most $m+1$ interpolation steps, and the final ceiling-enforcement arithmetic $32(m+1)^2 \cdot (m+1) = 32(m+1)^3$ are all correct as written; the two places where the writeup states weaker constants than it proves are deliberate slack, not errors. The single non-elementary ingredient (proportional interval splitting) is a correctly quoted weakening of the Stromquist-Woodall theorem, which I confirmed against the literature. The interpretation is exactly the source paper’s Conjecture 1.15 (arbitrary edge sets, existence of $c(m)$), checked verbatim against arXiv:1611.03196v1. Computationally, the combinatorial core passed 4000 randomized structural checks, brute force on 587 small instances never produced a deficit above $1/2$, and a full implementation of the pipeline produced valid matchings meeting every claimed intermediate and final bound. Literature searches found no prior resolution (the nearest work, Othman-Berger 2024, treats only $K_{\{n,n\}}$ partitions). Verdict: CONFIRMED — the conjecture is proved with $c(m) = 32(m+1)^3$; the authors’ suggested sharp value $m/2$ remains open.

References

- [1] R. Aharoni, N. Alon, E. Berger, M. Chudnovsky, D. Kotlar, M. Loebl and R. Ziv, Fair representation by independent sets, arXiv:1611.03196 (2016), Conjecture 1.15 (numbering of arXiv v1).
- [2] A. Othman and E. Berger, Almost fair perfect matchings in complete bipartite graphs, *Discrete Math.* 347 (2024).
- [3] W. Stromquist and D. R. Woodall, Sets on which several measures agree, *J. Math. Anal. Appl.* 108 (1985) 241–248.